Automatically Resolving Neutron Resonances with Machine Learning

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Background

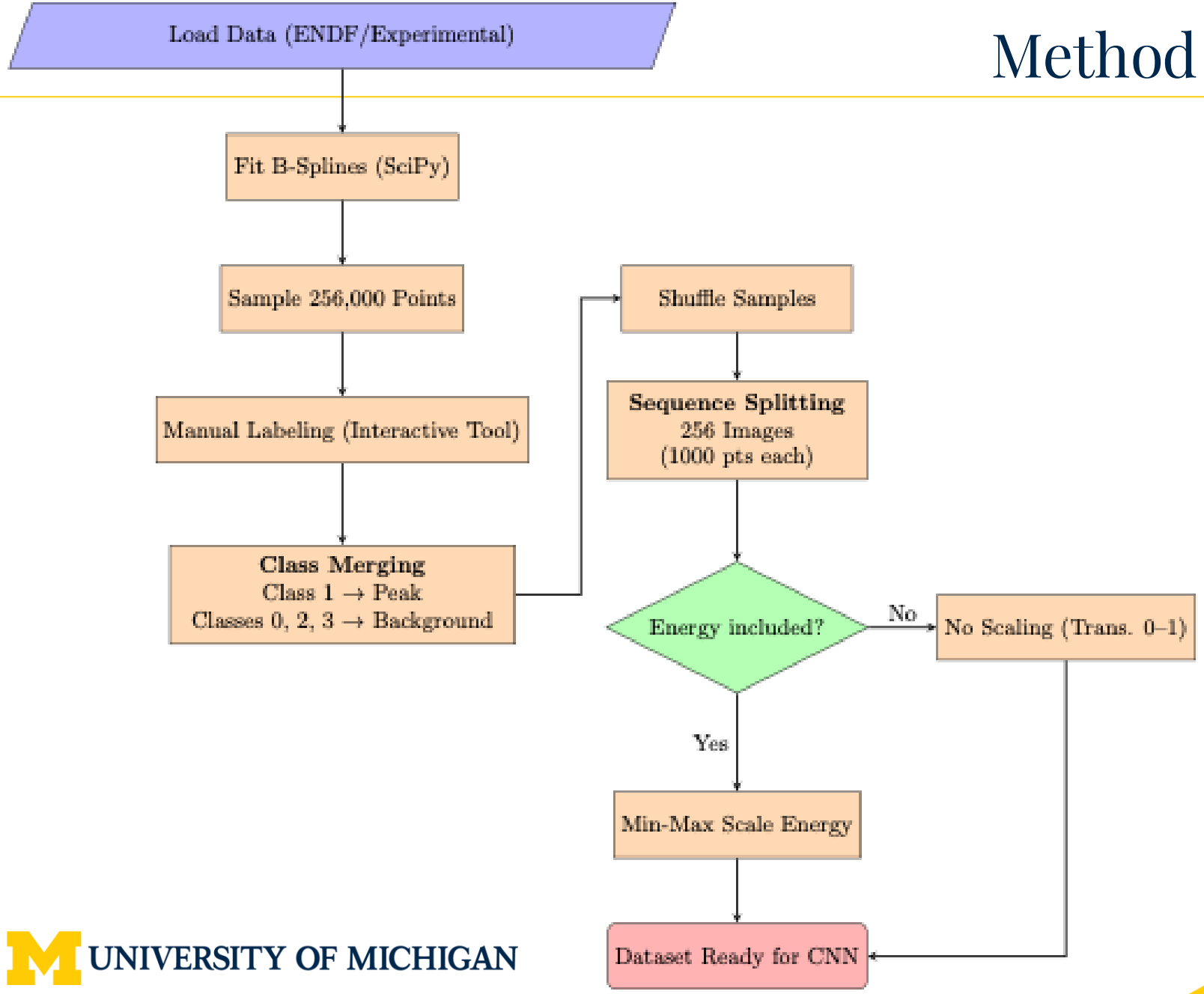
- Fitting neutron resonances where they can be experimentally resolved (in the resolved resonance region) is difficult and time-consuming.
- We can get raw transmission data using time-of-flight measurements from facilities such as the Los Alamos Neutron Science Center's (LANSCE) Device for Indirect Capture Experiments on Radionuclides (DICER).
- Traditionally, fitting these depends on prior evaluations in R-matrix codes, like SAMMY or REFIT, which necessarily skews new data toward old fits.

Goals

Automate the detection of the high frequency neutron resonances such that:

1. they do not depend on prior distributions that could overly skew results toward past evaluations, and
2. the process is streamlined to improve experimental post-processing efficiency.

Method - Preprocessing



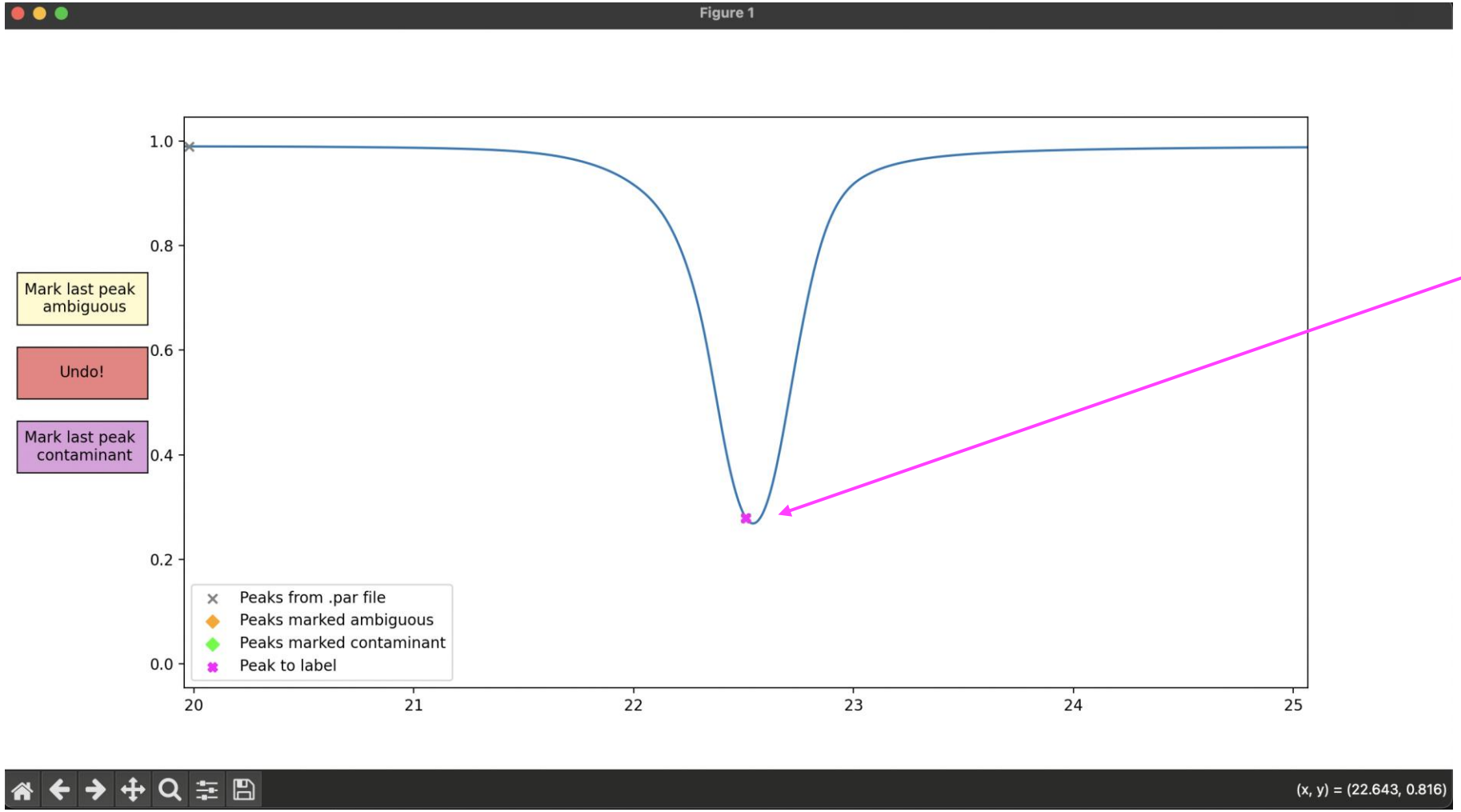
*Additional class labels: contaminant (2) or ambiguous (3)

**Shuffle (transmission, label) pairs so that they are discontinuous in energy

***Sequence lengths are arbitrary

****Transmission values range from 0-1, so no scaling needed

Method - Labeling



Points are loaded from a pre-existing parameter file (from SAMMY) as a guide.

Fig. 2: Python tool to map transmission array indices to peak/non-peak labels.

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Method - Labeling

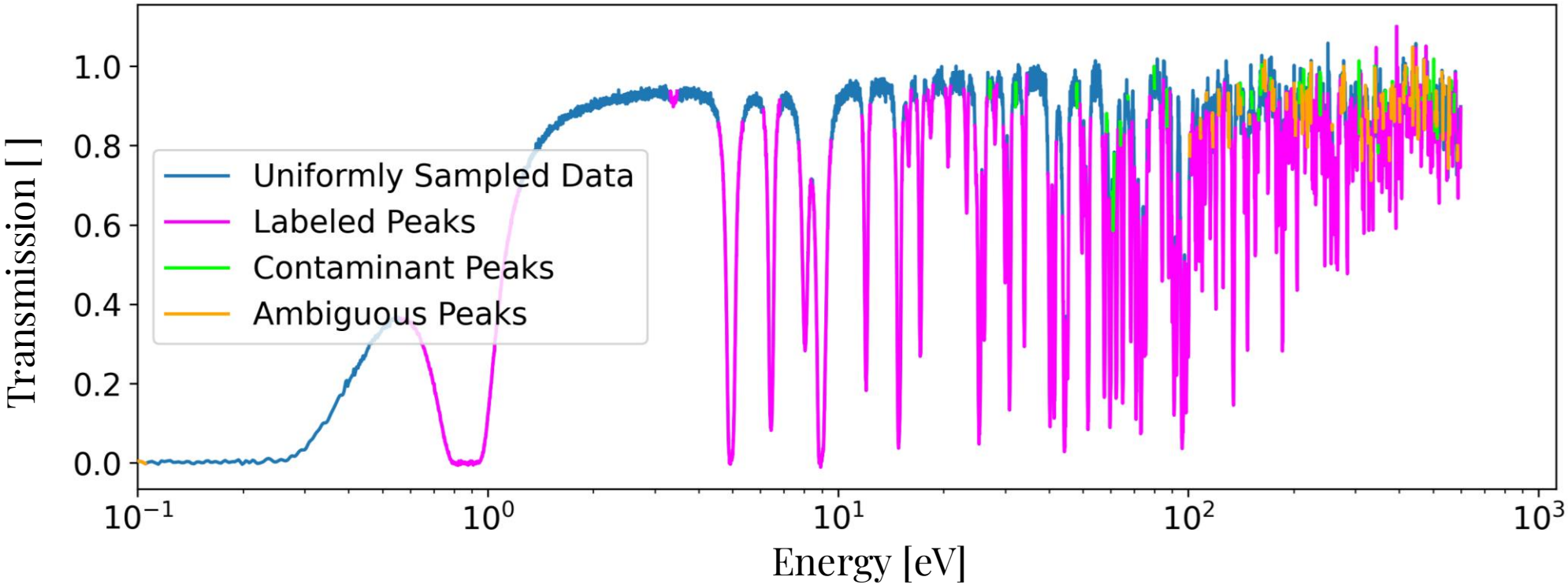


Fig. 3: Labeled Sm-149 experimental sample.

Method - Preprocessing

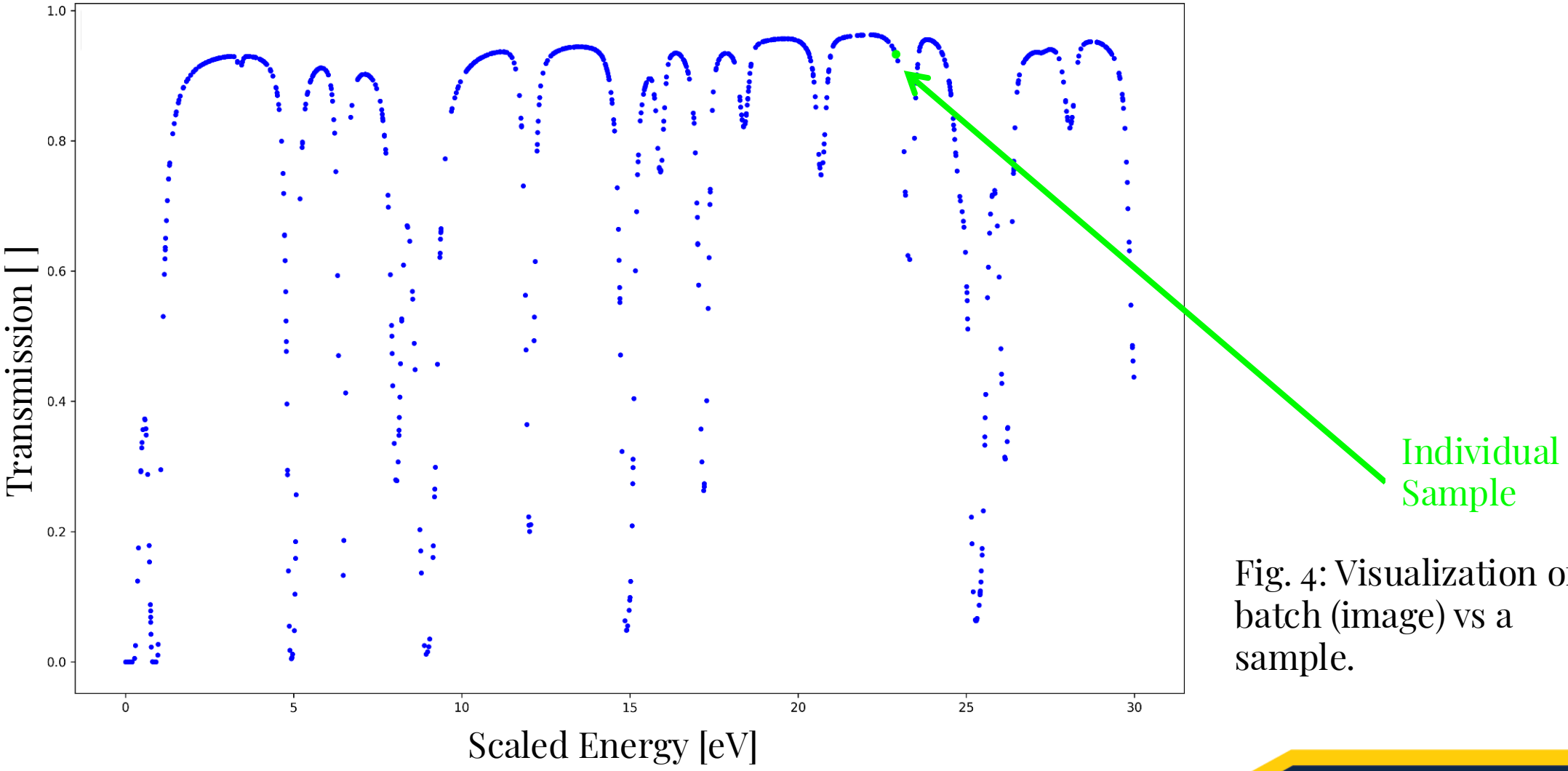
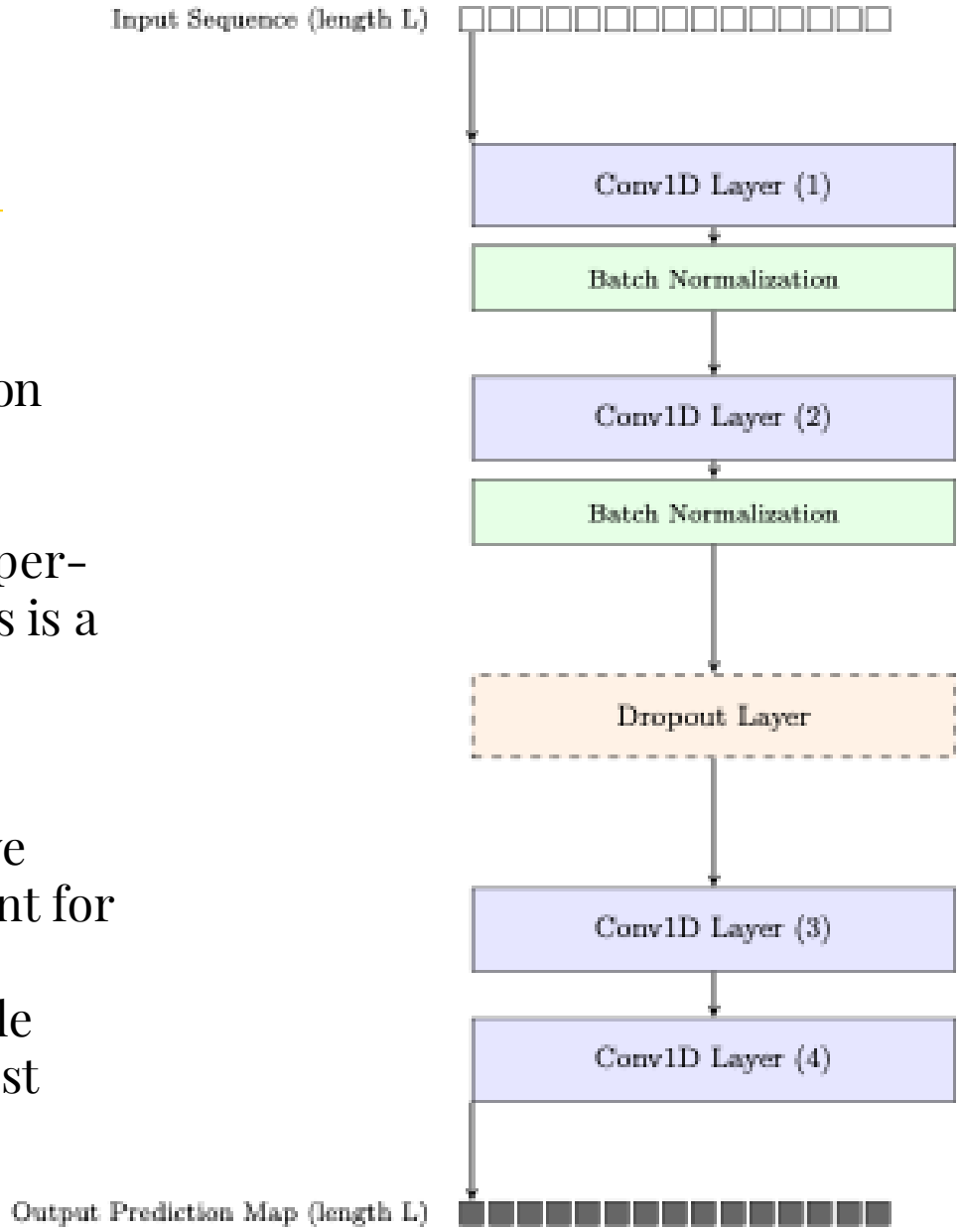


Fig. 4: Visualization of a batch (image) vs a sample.

Method – Modeling with a Fully Convolutional Network

- **Model architecture**
  - Two alternating Conv1d layers with batch normalization
  - Dropout layer
  - Two more Conv1d layers
  - **No fully connected layer** → Yields pixel-wise output (per-cell predictions instead of overall prediction, like, ”this is a zebra”)

- **Training**
  - Adam (optimizer) + BCE with Logits Loss and a positive weight of 1.33 (negative-positive sample ratio to account for class imbalance)
  - Exclude energy to make cross-spectra analysis possible
  - Five k-folds for single spectra analysis (80-20 train-test split)
  - 50 epochs



Method – Hyperparameter Tuning

Table 1. Best hyperparameters for the three datasets (two ENDF evaluations + one experimental).

Hyperparameter	Cs-133 Eval.	Sm-149 Eval.	Sm-149 Exp.
Filters	128	64	32
Kernel Sizes	[7, 9, 5, 7]	[3, 5, 5, 3]	[5, 5, 3, 9]
Dropout Rate	0.33	0.21	0.31
Batch Size	16	16	16
Initial Learning Rate	0.00517	0.00468	0.00264
Total Trainable Parameters	461,825	82,945	17,153

- Optimize F1 score over five k-folds using the Tree of Parzen Estimators algorithm (TPE) for 100 trials in Optuna.
- Only two epochs of training allowed during tuning

$$Precision = \frac{TP}{TP + FP} \qquad Recall = \frac{TP}{TP + FN} \qquad F1 = 2 \times \frac{precision \times recall}{precision + recall} \tag{1}$$

Eq 1. Show precision, recall, and F1 score, as used in hyperparameter tuning.

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Results – Cesium-133 Evaluation

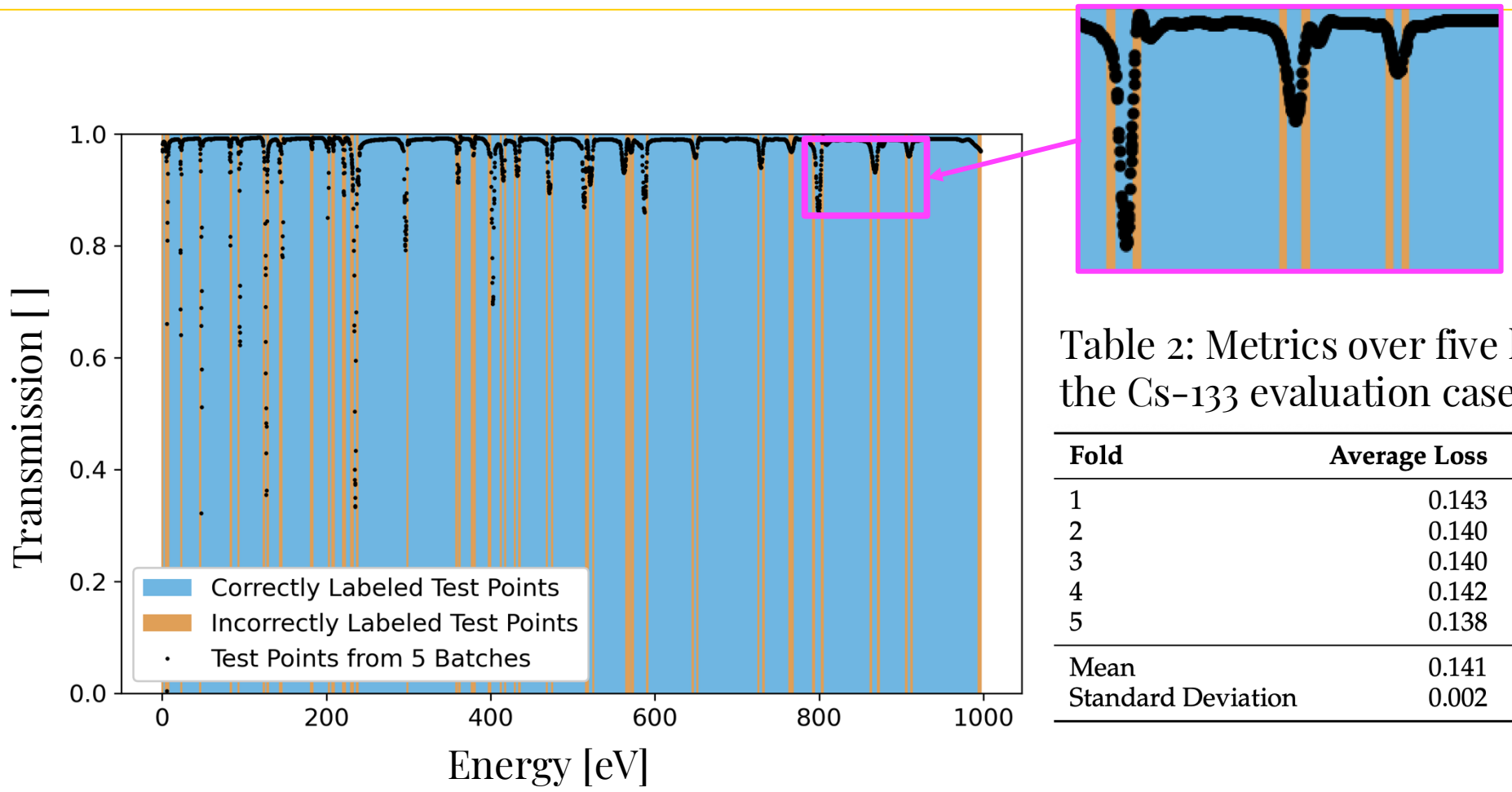
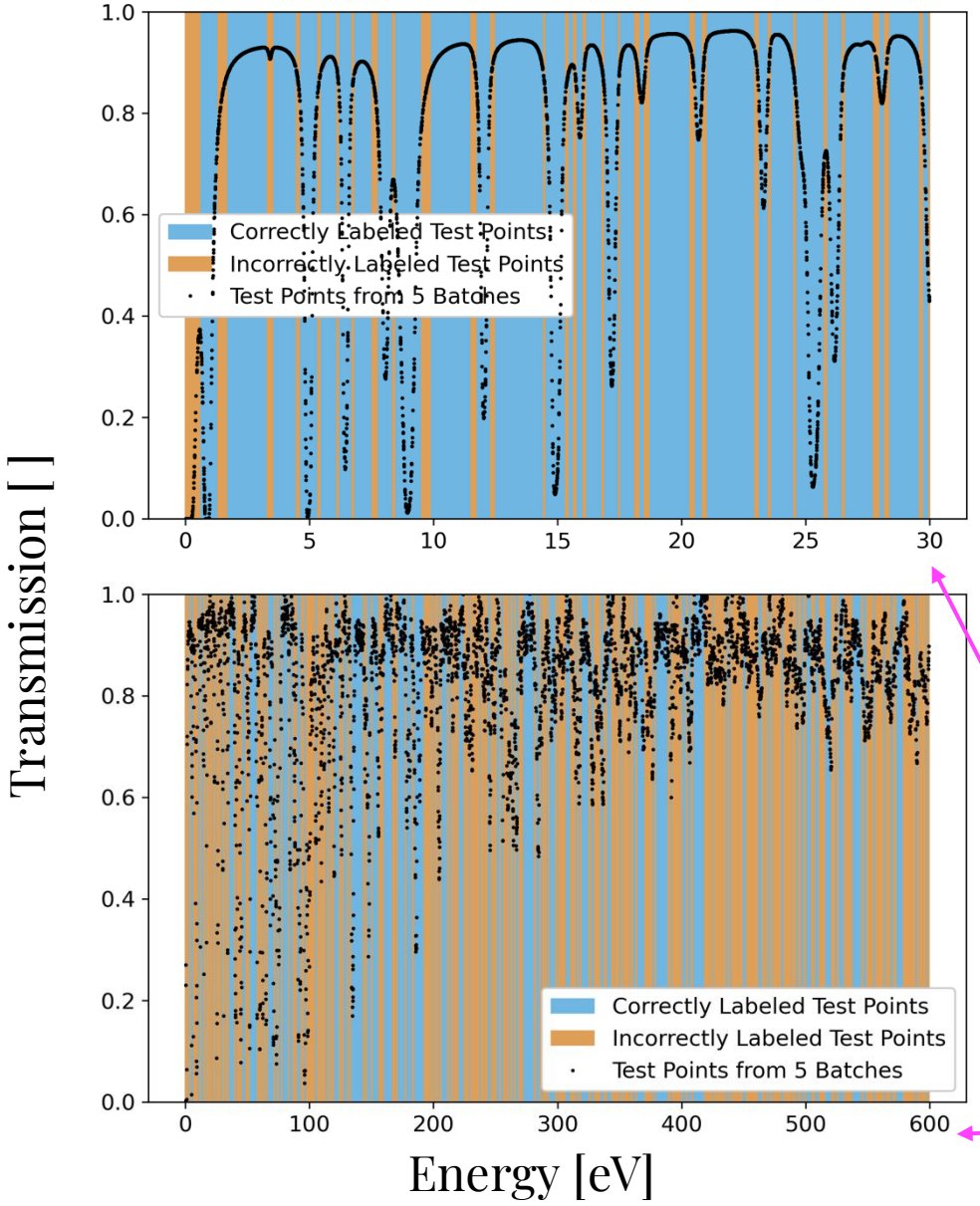


Table 2: Metrics over five k-folds of 50 epochs each for the Cs-133 evaluation case.

Fold	Average Loss	Accuracy	Precision	Recall	F1 Score
1	0.143	0.954	0.847	0.835	0.841
2	0.140	0.955	0.844	0.853	0.848
3	0.140	0.955	0.844	0.841	0.843
4	0.142	0.956	0.850	0.850	0.850
5	0.138	0.955	0.848	0.847	0.848
Mean	0.141	0.955	0.847	0.845	0.846
Standard Deviation	0.002	0.001	0.003	0.007	0.004

Fig. 3: Model predictions from the first k-fold model on five batches of unseen test data.



Results – Samarium-149 Evaluation & Experimental Cases

Table 3: Summary metrics over five k-folds of 50 epochs each for the Sm-149 evaluation and experimental cases

Case	Average Loss	Accuracy	Precision	Recall	F1 Score
Sm-149 Evaluation					
Mean	0.348	0.875	0.846	0.831	0.839
Standard Deviation	0.001	0.001	0.002	0.002	0.001
Sm-149 Experimental					
Mean	0.541	0.780	0.750	0.733	0.741
Standard Deviation	0.001	0.001	0.005	0.006	0.001

Energy range difference makes the evaluation dataset an easier problem

Fig. 4. Sm-149 evaluation (top) and experimental (bottom) model predictions from the first k-fold model on five batches of unseen test data.

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Results – Cross-Spectra Analysis

Difference due to the shape of the Sm-149 (deep) and Cs-133 (shallow) evaluation peaks.

Table 4: Metrics for training and testing on opposite evaluation data.

Case	Average Loss	Accuracy	Precision	Recall	F1 Score
Train-Cs, Test-Sm	0.597	0.845	0.874	0.708	0.782
Train-Sm, Test-Cs	0.371	0.878	0.966	0.171	0.291

Low recall and high precision means that the model has very few false positives but many false negatives.

*These results are promising given our data limitations, but it's clear that more work is needed to improve these results.

During cross-spectra training, fewer epochs yielded better results, indicating a **tendency to overfit. Again, increasing data should eliminate this problem.

Conclusions & Future Work

- As a feasibility study, these results are promising.
- Majority of error is coming from point-wise prediction framework, but the precise boundaries aren't critical for obtaining resonance parameters necessary to inform cross-section data.
- Worst performing case: Sm-149 experimental (expected)
- Possible routes for improvement:
  - More data!
  - More pre-processing and smoothing, especially for experimental datasets.
  - Other models, with less sensitivity to peak boundaries
  - Integration with existing R-matrix codes

Acknowledgments

The first author of this work is supported by the National Science Foundation's Graduate Research Fellowship Program (DGE-2241144). This work was supported by the U.S. Department of Energy through the Los Alamos National Laboratory. Los Alamos National Laboratory is operated by Triad National Security, LLC, for the National Nuclear Security Administration of the U.S. Department of Energy (Contract No. 89233218CNA000001). This was was also supported in part by the Nuclear Criticality Safety Program, funded and managed by the National Nuclear Security Administration for the Department of Energy.

References

[1] A. Stamatopoulos and et al. "New capability for neutron transmission measurements at LANSCE: The DICER instrument." *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment*, **volume 1025**, p. 166166 (2022).

[2] Nancy M. Larson. "Updated Users' Guide for SAMMY: Multilevel R-Matrix Fits to Neutron Data Using Bayes' Equations." Technical Report ORNL/TM-9179/R8, Oak Ridge National Laboratory (2008).

[3] M. Moxon. "REFIT: A least square fitting program for resonance analysis of neutron transmission data." (1979).

[4] Q.-K. Sun et al. "Enhancing reliability in photonuclear cross-section fitting with Bayesian neural networks." *Nuclear Science and Techniques*, **volume 36**(3), p. 52 (2025).

[5] R. Teelock-Gaya et al. "Prediction of Exotic (n,2n) Cross Sections Using a Regression Tree Machine Learning Algorithm." *Nuclear Science and Engineering*, **volume 0**(0), pp. 1–20. Publisher: Taylor & Francis _eprint: https://doi.org/10.1080/00295639.2025.2456414.

[6] B. Whewell et al. "Evaluating 239Pu(n,f) cross sections via machine learning using experimental data, covariances, and measurement features." *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment*, **volume 978**, p. 164305 (2020).

[7] Y. Ali Üncü et al. "Predicting (n,3n) nuclear reaction cross-sections using XGBoost and Leave-One-Out Cross-Validation." *Applied Radiation and Isotopes*, **volume 219**, p. 111714 (2025).

[8] C. Besnard-Vauterin et al. "Experimental data-driven modeling and prediction of (,n) cross-sections with physics-informed neural networks and gradient boosted decision trees." (2025).

[9] Z.-H. Hu et al. "Novel deep learning-based evaluation of neutron resonance cross sections." *Physics Letters B*, **volume 857**, p. 138978 (2024).

[10] A. Long. URL https://github.com/lanl/PLEIADES.

[11] M. I. Radaideh et al. "Modeling nuclear data uncertainties using deep neural networks." In *EPJ Web of Conferences*, volume 247, p. 15016. EDP Sciences (2021).

[12] N. A. W. Walton et al. "Automated Resonance Identification in Nuclear Data Evaluation." (2024).

[13] D. Kim and et al. "A novel hybrid CNN-transformer model for arrhythmia detection without R-peak identification using stockwell transform." *Scientific Reports*, **volume 15**(1), p. 7817 (2025). Publisher: Nature Publishing Group.

[14] A. D. Melnikov and et al. "Deep Learning for the Precise Peak Detection in High-Resolution LC–MS Data." (2020).

[15] D. Oh and et al. "CNN-Peaks: ChIP-Seq peak detection pipeline using convolutional neural networks that imitate human visual inspection." *Scientific Reports*, **volume 10**(1), p. 7933 (2020). Publisher: Nature Publishing Group.

[16] D. A. Brown and et al. "ENDF/B-VIII.0: The 8th Major Release of the Nuclear Reaction Data Library with CIELO-project Cross Sections, New Standards and Thermal Scattering Data." *Nuclear Data Sheets*, **volume 148**, pp. 1–142 (2018).